

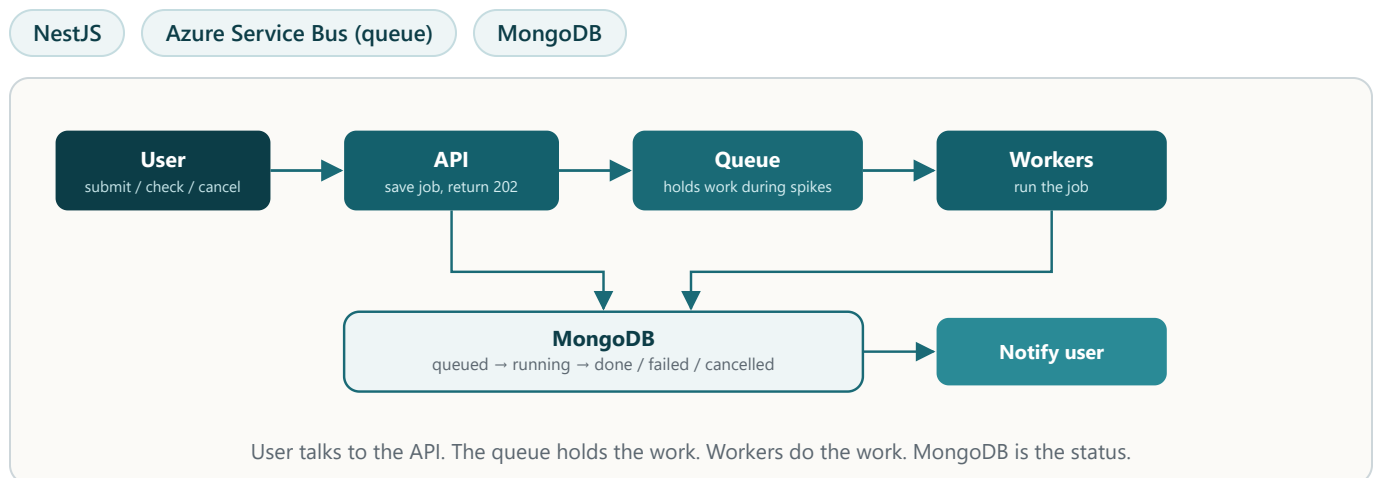
Async job processing

What they need

- Users submit jobs through an API
- Each job can take seconds to minutes
- Tell the user when it finishes or fails
- Let them check status while it runs
- Lots of jobs at once, including sudden spikes
- If a job fails, try it again
- Sometimes cancel a job that is already running

The solution

The API does not run the job. It saves the job, puts it on a queue, and returns right away. Workers pull jobs off the queue and run them. MongoDB holds the status so anyone can look it up. When the job finishes, we call the user's webhook.



How the pieces fit

- **API** — Accepts submit, status, and cancel. Saves the job, then puts it on the queue.
- **Queue (Service Bus)** — Holds jobs when more come in than workers can run.
- **Workers** — Take a job, run it, update status. Same NestJS codebase as the API, separate process.
- **MongoDB** — One record per job: status, attempts, cancel flag, webhook URL.

Q&A

What is the high-level architecture?

API, queue, workers, and a job record in MongoDB. The API never runs the long work. Status is always "read the job in MongoDB." Notify is a webhook when the job reaches done or failed.

What does each part do?

The API is the front door. The queue absorbs spikes. Workers do the work and write status back. MongoDB is the source of truth so cancel, retry, and "what's my job doing?" all look at the same record.

What are the hardest problems?

Cancel while it's running. You can't pull a job out of a worker's hands. Cancel sets a flag on the job. The worker checks that flag between steps. If it already finished, we say so. A job may finish the step it's on after cancel is requested.

Spikes. The queue holds the extra work. Add more workers when the queue grows. If one customer floods us, stop taking their new jobs until they catch up.

Retries. A failed job can be picked up more than once. The worker must be safe to run twice. Retry a few times with a wait in between, then mark it failed and tell the user.

Assumptions?

Users are already logged in. Jobs belong to a tenant. A job is just "run this payload." Cancel means "stop when you can," not "kill the process." Notification is a webhook they give us at submit time. We are on Azure: NestJS, Service Bus, MongoDB.